

Table 1: FactSales

Requirement 1 – Mandatory Fields Validation

Validate all mandatory fields before processing the sales data. The validation process must ensure that the following fields are not empty:

DateKey

ChannelKey

StoreKey

ProductKey

PromotionKey

CurrencyKey

UnitCost

UnitPrice

SalesQuantity

ReturnQuantity

TotalCost

SalesAmount

DiscountAmount

ReturnAmount

(Note: DiscountAmount and ReturnAmount were moved into the mandatory list after inspecting the actual staged data, which showed these fields are always populated — using 0 to represent "no discount/no return" rather than NULL.)

Any record that fails this validation must be rejected and logged with the appropriate rejection reason.

Requirement 2 – Numeric Values Validation

The validation process must reject any record containing one or more of the following:

SalesQuantity less than or equal to zero.

UnitCost less than zero.

UnitPrice less than or equal to zero.

ReturnQuantity less than zero.

TotalCost less than zero.

SalesAmount less than zero.

DiscountAmount less than zero.

ReturnAmount less than zero.

Requirement 3 – Business Rules Validation

The validation process must reject any record that violates one or more of the following rules:

DateKey must not be a future date.

ReturnQuantity must not be greater than SalesQuantity.

DiscountAmount must not be greater than SalesAmount.

Requirement 4 – Error Logging

Every rejected record must include: RecordID (SalesKey), SourceTable = 'FactSales', RejectionReason, ValidationTimestamp. This information must be available for auditing and troubleshooting purposes.

 Table 2: FactOnlineSales

Requirement 1 – Mandatory Fields Validation

DateKey

StoreKey

ProductKey

PromotionKey

CurrencyKey

CustomerKey

SalesOrderNumber

SalesOrderLineNumber

SalesQuantity

SalesAmount

ReturnQuantity

ReturnAmount

DiscountQuantity

DiscountAmount

TotalCost

UnitCost

UnitPrice

(All fields confirmed mandatory after inspecting the staged data — no NULLs found in any nullable-by-design column.)

Requirement 2 – Numeric Values Validation

SalesQuantity less than or equal to zero.

SalesAmount less than zero.

ReturnQuantity less than zero.

ReturnAmount less than zero.

DiscountQuantity less than zero.

DiscountAmount less than zero.

TotalCost less than zero.

UnitCost less than zero.

UnitPrice less than or equal to zero.

Requirement 3 – Business Rules Validation

DateKey must not be a future date.

ReturnQuantity must not be greater than SalesQuantity.

DiscountAmount must not be greater than SalesAmount.

The combination (SalesOrderNumber + SalesOrderLineNumber) must be unique.

Requirement 4 – Error Logging

RecordID (OnlineSalesKey), SourceTable = 'FactOnlineSales', RejectionReason,
ValidationTimestamp.

Table 3: FactInventory

Requirement 1 – Mandatory Fields Validation

DateKey

StoreKey

ProductKey

CurrencyKey

OnHandQuantity

OnOrderQuantity

SafetyStockQuantity

UnitCost

DaysInStock

MinDayInStock

MaxDayInStock

Aging

(All fields confirmed mandatory after inspecting the staged data — no NULLs found.)

Requirement 2 – Numeric Values Validation

OnHandQuantity less than zero.

OnOrderQuantity less than zero.

SafetyStockQuantity less than zero.

UnitCost less than or equal to zero.

DaysInStock less than zero.

Requirement 3 – Business Rules Validation

DateKey must not be a future date.

MinDayInStock must not be greater than MaxDayInStock.

Requirement 4 – Error Logging

RecordID (InventoryKey), SourceTable = 'FactInventory', RejectionReason, ValidationTimestamp.

 Table 4: FactExchangeRate

Requirement 1 – Mandatory Fields Validation

CurrencyKey

DateKey

AverageRate

EndOfDayRate

Requirement 2 – Numeric Values Validation

AverageRate less than or equal to zero.

EndOfDayRate less than or equal to zero.

Requirement 3 – Business Rules Validation

DateKey must not be a future date.

Requirement 4 – Error Logging

RecordID (ExchangeRateKey), SourceTable = 'FactExchangeRate', ErrorReason, ErrorDate.